

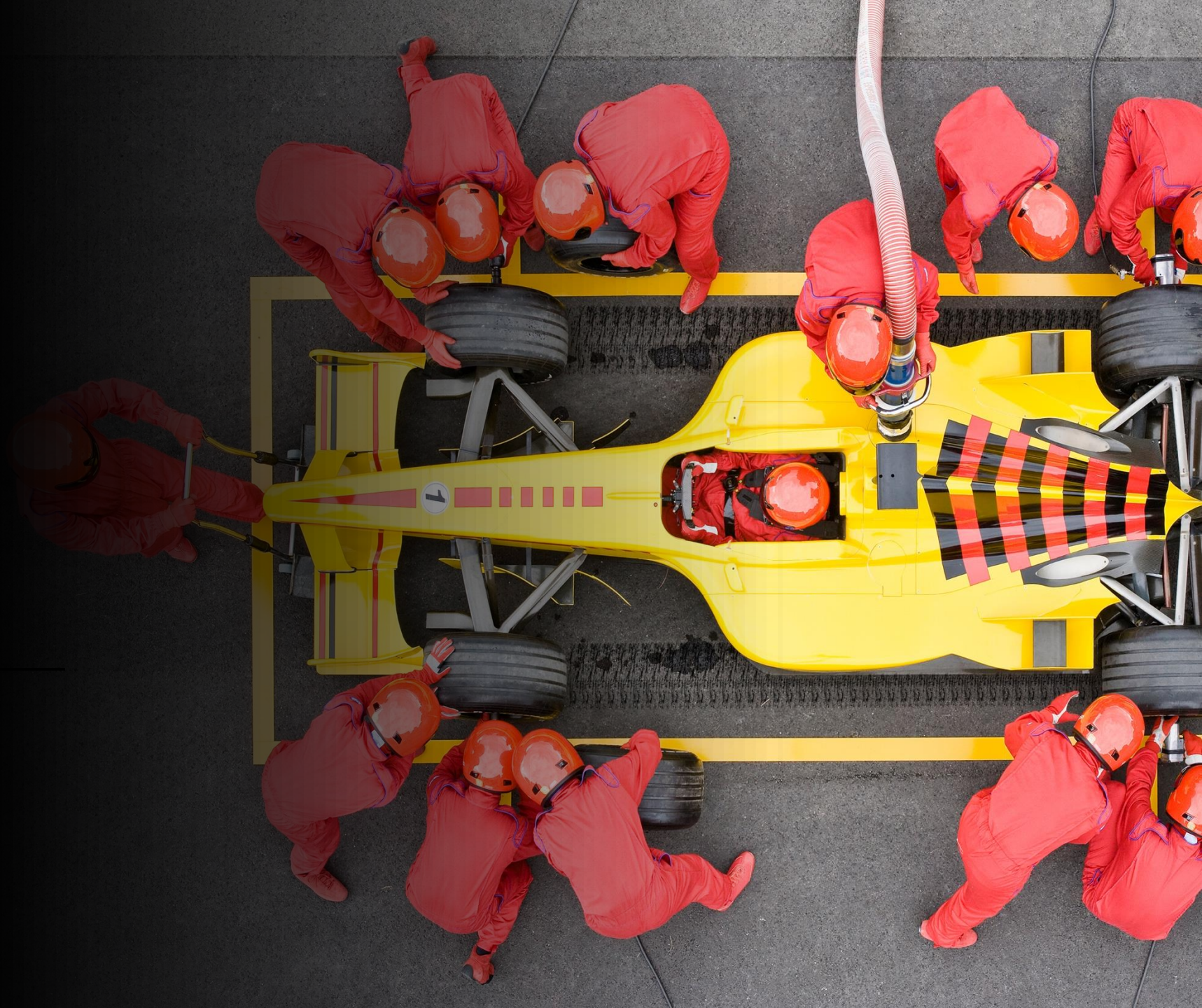
Predictive Analytics & Optimization in Formula 1 Fantasy

Project in Visualization - 4DT911

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The Challenge

Balancing immediate point production against volatile market dynamics under a hard cost cap.

Fantasy managers dilemma



Strict Roster Constraints



Dual-Objective Trade-off



Transfer Friction

The analytical challenge



The Average Trap



Multi-Modal Variables



Risk vs Reward

Data Architecture & Integration

- Unifying historical data with dynamic fantasy market economics.

CORE RELATIONAL DATA

Ergast F1 Historical Archive

1950–Present

- Driver, constructor, and race-level historical records.

MARKET & SCORING LAYER

F1 Fantasy Game Data

Dynamic Weekly Data

- Weekly pricing, ownership, and scoring dynamics.

SESSION CONTEXT

OpenF1 API

Telemetry Data

- Live weather, tire, and track-evolution telemetry.

Target Audience & Generalizability

- From fantasy managers to probabilistic modeling across sports and finance.

PRIMARY USERS

Target Audience



F1 Fantasy Managers

Casual and competitive players seeking data-driven transfers, budget maximization, and automated lineup optimization to climb league standings.



Sports Bettors & Wagering Analysts

Bettors looking for probabilistic finish projections, head-to-head match-up edges, and objective value lines against public odds.



Motorsport Fans & Data Enthusiasts

Track followers wanting interactive visual comparisons of car pace vs. driver skill and cost-efficiency benchmarks (PPM).

POSSIBLE EXTENSIONS

Generalizability & Cross-Domain Impact



Daily Fantasy Sports

The constrained combinatorial optimizer could translate to Premier League Fantasy and NFL Fantasy leagues.



Sports Wagering & Oddsmaking

Outcome distribution modeling (P10, P50, P90) can compute true fair-value odds, identify mispriced markets, and calculate risk margins.



Financial Portfolio Allocation

Balancing volatile, high-return assets with stable budget-enablers under a fixed ceiling mirrors classic portfolio construction.

Dataset 1: Historical Racing Archive (Ergast)

- Covers F1 history (1950–Present)

6,500,563 Data Points

14 Relational Tables

121 Features

Schema & Entity Architecture

- **Core Relational Backbone:** 14 relational tables linked by `raceId`, `driverId`, and `constructorId`.
- **High-Volume Tables:** `lap_times` (882k rows / 5.3M points), `results` (27.5k rows), `driver_standings` (35.7k rows), and `pit_stops` (22.7k rows).
- **Dimensional Types:** Quantitative coordinates (`lat`, `lng`, `alt`), ordinal finishing ranks (`position`), and categorical status codes (`Engine`, `Accident`, `Finished`).

Analysis

- 1,707 duplicate rows found in `lap_times.csv`, which is to be expected
- No missing values.

Dataset 2: Dynamic Game Economics (F1 Fantasy)

- Valuation metrics spanning 2022 to the present.

2022–Present

Weekly Temporal Step

Dynamic Valuations

187 Data Points

Schema & Entity Architecture

- Asset Universe:** Weekly valuations for 22 active drivers and 11 constructors.
- Core Dimensions:** Quantitative continuous pricing (`price_m`), total fantasy points, and ownership percentages (`selected_by_pct`).

Analysis

- Strict distribution in pricing: premium anchor assets (\$25M–\$35M) vs. budget enablers (\$5M–\$10M).
- High volatility in ownership percentage driven by recent race form and track matchups.
- Clean dataset with zero missing cells across active records.
- Standardized driver acronyms/codes (`VER`, `HAM`, `NOR`) to align with Ergast's historical `driverRef` schema.

Dataset 3: Real-Time Session Telemetry (OpenF1)

- Micro-temporal and track-level telemetry capturing active racing conditions.

Sub-Second Latency

16 Telemetry Dimensions

Micro-Spatial GPS

2,341,250 Data Points

Schema & Entity Architecture

- **Contextual Variables:** Track temperature, air temperature, humidity, rainfall indicators, and wind vectors.
- **Session Granularity:** Sector-by-sector time splits (`segments_sector_1/2/3`), speed trap readings, and pit-lane state flags (`is_pit_out_lap`).

Analysis

- Micro-temporal: sub-second telemetry synchronized against official session elapsed clocks.
- Micro-spatial: track coordinate mapping along physical race circuit sectors.
- Null cells resulting from pit exit out-laps prior to official timing line activation.